

# Gabee — Design & Brand Spec v0.1

Implementation-ready companion to **product-spec-v0.1 §15**. The product spec holds the *why*; this file holds the *how* — concrete tokens, components, and asset exports for the coding agent and any designer.

**Brand essence:** a warm, calm, trustworthy **robot bee** — a friendly guide, *not* an attention casino. One mascot does quadruple duty: brand mark, the letter **g**, app icon, and in-app companion.

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## 1. Logo & wordmark

**Concept** — the robot bee is the letter "g" in **gabee**.

- **Wordmark:** "abee" set in **Mulish 800**; the "g" is a bespoke double-storey bee following the closed-loop "g" of **Open Sans** (geometric reference only — not a loaded webfont).
- **Construction:**
  - Head diameter = lowercase "a", aligned on the x-height (the g's upper bowl).
  - Abdomen = **same size as the head**, sitting as the descender below the baseline (the g's lower loop).
  - Two **antennae** above the head with **cyan ball tips**.
  - Two dark **stripes** on the abdomen.
- **Wordmark has no wings** (keeps the lockup compact).
- **Clear space:** keep a margin  $\geq$  the bee's head diameter on all sides.
- **Minimum width:** wordmark  $\geq$  96px; below that, use the icon (mascot) instead.

### Do / Don't

- Do keep the body honey, eyes/antenna-tips cyan, visor + outlines ink.
  - Don't recolour the body, stretch the lockup, add a sad/angry face, or place the ochre module colour directly behind the logo (too close to honey).
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## 2. Mascot, standalone logo & app icon

- **Standalone logo / mascot / app icon** = full bee **with wings** (light-cyan #BBEAF2 ellipses behind the body).
  - **App icon:** winged bee centred on an ink #20242E rounded-square (radius --radius-x1).
  - **Favicon /  $\leq$  16px:** use a **wingless** variant so the silhouette stays legible.
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### 3. Expression system

The **logo face is fixed and robotic** (dark visor + two neutral cyan dots) on all brand surfaces. **In-app, the visor is a screen:** body/wings/antennae/stripes never move — only the cyan content of the visor changes.

| Token         | Situation            | Visor expression                                                                                   | Triggered by (analytics event, ref product §9.3) |
|---------------|----------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------|
| bee/idle      | Idle / home          | two calm dots<br> | default                                          |
| bee/correct   | Correct answer       | smiling eyes<br>  | answer correct                                   |
| bee/celebrate | Level complete       | star eyes         | code_level_solved / level cleared                |
| bee/encourage | After a wrong answer | supportive <b>wink</b>                                                                             | answer wrong                                     |
| bee/focus     | Question on screen   | dots looking up                                                                                    | question shown                                   |

- Wrong-answer state is encouraging — never sad or shaming** (ref product §6.3).
- Motion:** cross-fade/morph the visor content over `--motion-base` (200ms), ease-out. Celebration may be slightly longer but **must not flash or strobe**. Respect `prefers-reduced-motion` (swap to instant state change).
- Extendable later: idle blink, screen pulse while the voiceover speaks — without touching the body.

### 4. Colour system

Colour is a **wayfinding** device, not a symbol. Each module keeps its colour permanently; tiles **always pair colour with an icon + label**, so colour is reinforcement, never the only cue (covers colour-blind users).

#### 4.1 Module colours

| Token            | Module  | Colour  | Hex                                                                                           | Text on it |
|------------------|---------|---------|-----------------------------------------------------------------------------------------------|------------|
| --module-numbers | Numbers | Blue    |  #1F6FEB | white      |
| --module-words   | Words   | Magenta |  #D6336C | white      |

| Token                | Module      | Colour     | Hex                                                                                         | Text on it |
|----------------------|-------------|------------|---------------------------------------------------------------------------------------------|------------|
| --module-keyboard    | Keyboard    | Ochre      |  #C99A0E | ink        |
| --module-code        | Code        | Violet     |  #7B2FF7 | white      |
| --module-translation | Translation | Terracotta |  #C75D28 | white      |

## 4.2 Brand & interface

| Token           | Role                    | Hex                                                                                          | Text on it |
|-----------------|-------------------------|----------------------------------------------------------------------------------------------|------------|
| --color-brand   | Brand (honey)           |  #FFB400   | ink        |
| --color-ink     | Text / visor / outlines |  #20242E   | white      |
| --color-accent  | Accent / bee eyes       |  #2BD4E6   | ink        |
| --color-wing    | Wings                   |  #BBEAF2   | ink        |
| --surface       | Page surface            |  #FFFFFF   | ink        |
| --surface-muted | Muted panel             |  #F7F8FA  | ink        |
| --border        | Hairline / divider      |  #E6E8EE | —          |
| --text          | Primary text            |  #20242E | —          |
| --text-muted    | Secondary text          |  #6B7280 | —          |

## 4.3 Feedback (reserved — never a module colour)

| Token              | Meaning   | Hex                                                                                           |
|--------------------|-----------|-----------------------------------------------------------------------------------------------|
| --feedback-correct | Correct   |  #3F7A2E |
| --feedback-retry   | Try again |  #E5322B |

## 4.4 Usage rules & watch-points

- **Red & green are feedback only** — never module identity. Keeps right/wrong unambiguous and avoids a red+green colour-blind clash between modules.
- **Honey is the brand.** Ochre (Keyboard) is the closest hue; verify legibility next to the bee and desaturate toward olive (~ #B8901C)) if it muddies.
- **Cyan is the accent/eyes**, not a module colour.
- The "correct" celebration may lean on **honey + the star-eye bee** rather than green alone.

- **Text-on-colour:** use **ink** on honey, ochre, cyan; **white** on blue, magenta, violet, terracotta, green, red.

## 5. Typography

- **Mulish** — primary typeface: wordmark ("abee", 800) **and all UI.** Rounded, friendly, legible at large sizes.
- **Open Sans** — reference for the bespoke "g" glyph only (not loaded as a webfont).
- Weights in use: 400 (body), 600 (labels/UI), 700 (subheads), 800 (display/wordmark).

**Type scale** (kid-facing minimum body = 18px; generous line-height):

| Token                     | Use              | Size / line-height | Weight |
|---------------------------|------------------|--------------------|--------|
| <code>text-display</code> | Hero / wordmark  | 56 / 60            | 800    |
| <code>text-h1</code>      | Screen title     | 36 / 44            | 800    |
| <code>text-h2</code>      | Section          | 28 / 36            | 700    |
| <code>text-h3</code>      | Card title       | 22 / 30            | 700    |
| <code>text-body-lg</code> | Kid prompts      | 20 / 30            | 600    |
| <code>text-body</code>    | Body             | 18 / 28            | 400    |
| <code>text-label</code>   | Buttons / labels | 16 / 24            | 600    |
| <code>text-caption</code> | Captions         | 14 / 20            | 600    |

In any UI numerals, ensure `1 / l / I` and `0 / o` stay distinguishable.

## 6. Design tokens

### 6.1 Radius (rounded, generous)

| Token                    | Value | Use                 |
|--------------------------|-------|---------------------|
| <code>--radius-sm</code> | 8px   | chips, small inputs |
| <code>--radius-md</code> | 14px  | inputs, small cards |
| <code>--radius-lg</code> | 20px  | buttons, cards      |

| Token                      | Value  | Use                    |
|----------------------------|--------|------------------------|
| <code>--radius-xl</code>   | 28px   | module tiles, app icon |
| <code>--radius-pill</code> | 9999px | pills, the bee's visor |

### 6.2 Spacing (4px base)

`--space-1:4` · `--space-2:8` · `--space-3:12` · `--space-4:16` · `--space-5:20` · `--space-6:24`  
 · `--space-8:32` · `--space-10:40` · `--space-12:48` · `--space-16:64`

### 6.3 Elevation (soft, low-contrast)

| Token                    | Value                                       |
|--------------------------|---------------------------------------------|
| <code>--shadow-sm</code> | <code>0 1px 2px rgba(32,36,46,.06)</code>   |
| <code>--shadow-md</code> | <code>0 6px 16px rgba(32,36,46,.10)</code>  |
| <code>--shadow-lg</code> | <code>0 12px 32px rgba(32,36,46,.12)</code> |

### 6.4 Motion

| Token                      | Value                                 | Use                           |
|----------------------------|---------------------------------------|-------------------------------|
| <code>--motion-fast</code> | 120ms                                 | hovers, taps                  |
| <code>--motion-base</code> | 200ms                                 | expression morph, transitions |
| <code>--motion-slow</code> | 320ms                                 | celebration enter             |
| <code>--easing</code>      | <code>cubic-bezier(.2,.8,.2,1)</code> | default ease-out              |

Always honour `prefers-reduced-motion`: no parallax, no looping attention-grabbers, no flashing.

### 6.5 CSS custom properties (drop-in `:root`)

|     |
|-----|
| CSS |
|-----|

```

:root {
  /* brand & ui */
  --color-brand:#FFB400; --color-ink:#20242E; --color-accent:#2BD4E6; --color-wing:#
  --surface:#FFFFFF; --surface-muted:#F7F8FA; --border:#E6E8EE; --text:#20242E; --te
  /* modules */
  --module-numbers:#1F6FEB; --module-words:#D6336C; --module-keyboard:#C99A0E;
  --module-code:#7B2FF7; --module-translation:#C75D28;
  /* feedback */
  --feedback-correct:#3F7A2E; --feedback-retry:#E5322B;
  /* radius */
  --radius-sm:8px; --radius-md:14px; --radius-lg:20px; --radius-xl:28px; --radius-pi
  /* elevation */
  --shadow-sm:0 1px 2px rgba(32,36,46,.06);
  --shadow-md:0 6px 16px rgba(32,36,46,.10);
  --shadow-lg:0 12px 32px rgba(32,36,46,.12);
  /* motion */
  --motion-fast:120ms; --motion-base:200ms; --motion-slow:320ms;
  --easing:cubic-bezier(.2,.8,.2,1);
}

```

## 6.6 Tailwind theme mapping (excerpt)

```

js
// tailwind.config - theme.extend
colors: {
  brand:'#FFB400', ink:'#20242E', accent:'#2BD4E6', wing:'#BBEAF2',
  module:{ numbers:'#1F6FEB', words:'#D6336C', keyboard:'#C99A0E', code:'#7B2FF7', t
  feedback:{ correct:'#3F7A2E', retry:'#E5322B' },
},
borderRadius:{ sm:'8px', md:'14px', lg:'20px', xl:'28px' },
fontFamily:{ sans:['Mulish','system-ui','sans-serif'] },

```

## 7. Component guidance

- **Module tile** — `--radius-xl`, large, themed with its module colour; always shows **icon** + **label** + (optionally) the bee. One module colour per screen context.
- **Primary button** — `--radius-lg`, large target, honey or the active module colour, ink/white text per the contrast rule, `--shadow-md`.
- **Bee companion** — persistent guide; map its expression to events per §3. Sits near the question/feedback area; never blocks content.
- **Feedback**
  - *Correct:* `--feedback-correct` accent + **honey** + **star-eye bee**, brief positive copy.

- *Try again:* `--feedback-retry` accent + **wink bee**, gentle copy, **no penalty, no shaming** (ref product §6.3).
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## 8. UI principles

- **Rounded everything**; soft shapes echo the bee.
  - **Large targets, calm spacing**, one primary action per screen.
  - **One module colour per context**; bee + honey + ink stay constant for cohesion.
  - **No dark patterns** — no FOMO, no streak-guilt, no red-dot nagging (ref product §6.3); gamification stays content-neutral (ref product §6.1).
  - **Desktop-first** (keyboard + mouse are learning objectives), but rounded/large enough to stay touch-friendly.
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## 9. Accessibility

- **Contrast**: follow the text-on-colour rule (§4.4); target WCAG AA for body text. Ochre & cyan require ink text.
  - **Never colour-only**: every module/feedback state also carries an icon + label/copy.
  - **Target size**:  $\geq 44\text{px}$  minimum;  $\geq 56\text{px}$  **recommended** for this age group.
  - **Motion**: honour `prefers-reduced-motion`; no flashing/strobing.
  - **Legible numerals/letters** in UI (`1/1/I`, `0/0`).
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## 10. Asset export checklist

| Asset                    | Format                     | Notes                             |
|--------------------------|----------------------------|-----------------------------------|
| Wordmark                 | SVG                        | no wings; ink + honey + cyan      |
| Mascot / standalone logo | SVG                        | <b>with wings</b>                 |
| App icon                 | SVG + PNG 1024 / 512 / 192 | winged bee on ink rounded-square  |
| Apple touch icon         | PNG 180                    | winged, ink bg                    |
| Favicon                  | PNG/ICO 32 / 16            | <b>wingless</b> variant           |
| Splash screen            | SVG/PNG                    | bee + wordmark, surface or ink bg |
| Expression sheet         | SVG (per state) or sprite  | 5 visor states from §3            |

| Asset           | Format | Notes          |
|-----------------|--------|----------------|
| Colour swatches | —      | tokens from §4 |
|                 |        |                |
|                 |        |                |

*Source of truth for narrative & rationale: product-spec-v0.1 §15. Keep both files in sync if the identity changes.*